

Ziwei Guo

Daly City, CA — (223)-533-1053 — gzw19914760905@outlook.com — LinkedIn — GitHub — Website

Vanderbilt University, Nashville, TN

Aug 2024 – May 2026

M.S. in Computer Science, GPA: 3.75/4.00

Dickinson College, Carlisle, PA

Aug 2020 – May 2024

B.S. in Mathematics & Data Analytics

Technical Skills

Languages: Python, Go, SQL, TypeScript/JavaScript

Market Data & Research Systems: Nasdaq ITCH 5.0, MoldUDP64, binary protocols, order-book reconstruction, feed sequencing, point-in-time features, PyArrow, Parquet, DuckDB

Machine Learning & Agents: NumPy, pandas, scikit-learn, model evaluation, RAG, tool-calling agents, MCP, Playwright

Correctness & Testing: Pytest, Hypothesis, property-based and differential testing, state invariants, canonical hashing, fault injection, snapshot recovery

Systems & Infrastructure: Linux/Unix, Kubernetes, Docker, Kafka, Redis, Vault, AWS, CI/CD, event-driven systems, bounded queues, backpressure

Backend & Observability: FastAPI, Flask, PostgreSQL, REST APIs, WebSockets, Prometheus, Grafana, PromQL, profiling, telemetry, alerting

Experience

Tesla, Inc. — **Factory Software**, Fremont, CA

May 2025 – Aug 2025

Software Engineer Intern

- Built a production observability and diagnostics system deployed across Fremont, Reno, and Buffalo that translated operational questions into structured Prometheus and Grafana KPI checks over WebSocket and REST APIs; reduced incident triage time by **30%+** and saved **10+ engineer-hours per week**.
- Replaced static YAML configuration with **Grafana API-driven discovery** of PromQL queries, thresholds, and dashboard metadata from **5 production dashboards**, enabling runtime updates across **3 factory sites** without code changes.
- Engineered Kafka TLS certificate monitoring and automated rotation using cert-manager, Vault, Prometheus, and Kubernetes; validated secret consistency and safely rolled **5 production services** to prevent credential-expiration outages.
- Built a multi-cluster service-account rotation system with database validation, deduplication, retries, delayed-rollout safeguards, rolling restarts, and alerting for **12+ production services**.

Selected Projects

Market Data Integrity & Replay Platform — Nasdaq ITCH 5.0 / MoldUDP64

2026

- Built an official-spec Python market-data platform supporting all **23 Nasdaq TotalView-ITCH 5.0 message types** and MoldUDP64 transport semantics, including strict binary validation, packet sequencing, exact gap detection, duplicate suppression, A/B feed arbitration, and deterministic archive recovery.
- Scanned a full-day Nasdaq ITCH file containing **423,285,709 messages** and **12.95 GB** of uncompressed data in bounded memory; produced point-in-time feature, trade, and order-lifecycle Parquet datasets with DuckDB querying, event-level lineage, and identical logical hashes across repeated replays.
- Independently verified all supported messages across the optimized parser, a separately implemented audit parser, and **itchfeed 1.0.7** with **zero differences**; validated reference and optimized books through property-based differential tests and canonical checkpoint hashes.
- Executed **1,000 deterministic chaos scenarios** covering loss, duplication, reordering, corruption, snapshot interruption, and storage faults, yielding **900 clean-equivalent recoveries** and **100 expected typed failures** with reproducible evidence.
- Profiled a real **20-million-message segment** and reduced runtime from **596.775 to 256.235 seconds**, a **2.33× speedup** without changing logical hashes; completed **215 passing tests**, strict mypy and Ruff validation, and **93.44% test coverage**.

AgentCert — Deterministic Agent Execution and Reliability Platform

2026

- Built and deployed a Python and TypeScript platform for deterministic agent evaluation, policy enforcement, fault injection, and post-action verification; published as **agentcert@0.6.3** with CLI/CI integration and **50+ automated tests**.
- Designed stateful allow, approval, and block workflows with replay protection, idempotency, signed records, outcome probes, failure classification, and reproducible audit evidence.
- Delivered a multi-tenant PostgreSQL service with RBAC, API keys, incident tracking, webhook alerts, scope fingerprinting, and hash-linked JSON, JUnit, HTML, and Markdown outputs.

GridQueue Agent — Public Infrastructure Data and Reproducible Risk Analysis

2026

- Built a FastAPI, DuckDB, and Next.js pipeline over real ERCOT and LBNL datasets that normalized monthly files, resolved entities, detected **210 material queue events**, and generated a **2,206-row entity-resolution audit** with source traces, confidence signals, file hashes, and reproducibility metadata.

Research, Publications & Awards

Independent Research — Two-Dimensional Jacobian Conjecture

2026 – Present

- Studying weighted degenerations, associated-graded maps, and valuation-based reductions for plane Keller maps; developed exact symbolic-computation certificates and counterexamples ruling out several naive one-weight reduction routes. Manuscript in preparation.

Selected Publications: First author of “Improving Model-Free Control Algorithms” (*Mathematics*, 2024), “Research on Big Data Analysis and Prediction” (2023), and “Optimal Control Design for Distributed Drive Electric Vehicles” (2023).

Kaggle Competitions: 2 Silver Medals (OTTO, G2Net); 1 Bronze Medal (RSNA).